

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**1st Semester of B. Pharm. Examination****University Theory Examination December 2016****PH116 Pharmaceutical Biology-I****Date: 02.01.2017, Monday****Time: 10:00 a.m. to 1:00 p.m.****Maximum Marks: 80****Instructions:**

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I**Q 1** Attempt all questions. Each question is of one mark.**20**

1. In the cell cycle, DNA synthesis takes place during
[A] G₁ Phase
[B] G₂ Phase
[C] S Phase
[D] Prophase
2. Myasthenia gravis is the disease of
[A] Muscle
[B] Brain
[C] Heart
[D] Bones
3. The layer of connective tissue that separates the muscle tissue into small sections is called
[A] Aponeuroses
[B] Epimysium
[C] Perimysium
[D] Endomysium
4. Which of these statements is correct regarding muscle contraction?
[A] All motor units act together.
[B] Muscle contraction continues for long periods after nervous stimulation ceases.
[C] The crossbridges bind to the actin and shorten the sarcomeres.
[D] Dystrophin is not needed to strengthen the contracting muscle cell.
5. Anatomy means the study of
[A] Physiology
[B] Morphology
[C] Cell functions
[D] Human functions

6. The phycologist is the person who is involved in the study of
[A] Fungi
[B] Algae
[C] Virus
[D] Protozoa
7. Which of the following is cartilage connective tissue?
[A] Hyaline
[B] Blood tissue
[C] Adipose tissue
[D] All of the above
8. The oval window is located at the base of the
[A] Scala vestibule
[B] Auditory tube
[C] Cochlear duct
[D] Scala tympani
9. Which of these voluntary muscles are not striated?
[A] Cardiac
[B] Smooth
[C] Skeletal
[D] Facial
10. How many chromosomes are there in human cell?
[A] 45
[B] 47
[C] 46
[D] 48
11. Special staining techniques are used to identify
[A] Flagella
[B] Capsule
[C] Endospore
[D] All of the above
12. Mordant used in gram staining is
[A] Crystal Violet
[B] Iodine
[C] Alcohol
[D] Nigrosin
13. The shape of the epithelial tissue is _____
[A] Squamous
[B] Cuboidal
[C] Columnar
[D] All of the above

14. When a nervous impulse travels from a neuron to a muscle cell, what happens next?
 [A] The impulse travels over the sarcolemma in all directions
 [B] Calcium is released from the sarcoplasmic reticulum
 [C] Linkages form between the actin and myosin
 [D] Acetylcholine is decomposed by acetylcholinesterase
15. Night blindness can be treated with
 [A] Vitamin C
 [B] Vitamin D
 [C] Vitamin K
 [D] Vitamin A
16. Proprioceptive sensations are a type of _____.
 [A] Somatic Senses
 [B] Special Senses
 [C] Visceral Senses
 [D] Thermal Senses
17. Receptors for general senses provide information for all of the following except
 [A] Temperature
 [B] Pain
 [C] Vision
 [D] Touch
18. Taste Buds are found in the _____.
 [A] Olfactory epithelium
 [B] Papillae
 [C] Gustatory epithelium
 [D] Epiglottis
19. Incubation is a process which _____ of microbes.
 [A] Inhibits the growth
 [B] Promotes the growth
 [C] Maintain the growth
 [D] None of the above
20. The perception of color comes from all of the following except
 [A] Red cones
 [B] Yellow cones
 [C] Green cones
 [D] Blue cones

SECTION - II

Q 2 Attempt any **TWO** of the following

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|----------|---|-----------|
| A | Define the epithelial tissue and describe the functions of epithelial tissue. | 05 |
| B | Describe the structure of the olfactory receptors and physiology of Olfaction in brief. | 05 |
| C | Describe the following terms:
(i) Nitrogen fixation by microbes (ii) Bioremediation | 05 |

- Q 3** Attempt any TWO of the following
- A** Define Microbes. What is the application of microbes in pharmaceutical industry? **05**
 - B** Discuss the functions of skin and explain its utility in the process of 'Thermoregulation'. **05**
 - C** Recognize the connective tissue which is in liquid form and explain in brief. **05**

SECTION – III

- Q 4** Attempt any FOUR of the following
- A** Differentiate between eukaryotic and prokaryotic cells. **05**
 - B** Explain in detail the physiology of 'Muscle contraction'. **05**
 - C** Draw the labelled diagram of cell and write a note on nucleus. **05**
 - D** Discuss the physiology of hearing in detail. **05**
 - E** Classify muscular tissue. Compare and differentiate between these different types of muscles. **05**
 - F** Justify: The basic dyes stain the bacteria while acidic dyes stain the background during monochrome staining. **05**
- Q 5** Attempt any FOUR of the following
- A** Comment: "Mitochondria is the powerhouse of the cell." **05**
 - B** Draw a neat and labelled diagram of cross section of human eye. **05**
 - C** Write a note on neuromuscular junction with suitable diagram. **05**
 - D** Enumerate the different staining techniques for identification of microorganisms and describe any one method in detail. **05**
 - E** Explain the process of 'cell division' with its significance. **05**
 - F** Write in detail about the different transport mechanisms across cell membrane with suitable examples. **05**
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